

MASTER-COPY Skilled Worker Know-how

Document ID: MC-PDF-SS6G-ROOT-001 / Demo PDF for RAG parsing and ChromaDB indexing

Material	stainless	Position	6G
Process	root_pass	Knowledge type	expert_tip
Current range	82-88A	Backing gas	30-35 CFH
Test keyword	MC-PDF-SS6G-ROOT-001	Source video	demo_stainless_6g_root

1. Pre-check before the stainless 6G root pass

Before welding stainless 6G root pass, check purge leakage before changing current. Verify that both pipe ends are sealed, the purge hose is stable, and oxygen is below 50 ppm if an oxygen meter is available.

Keep the root gap near 2.4 mm and the land near 1.5 mm. If the gap opens at the 6 o clock quadrant, the puddle will sag and suck-back becomes likely.

Use 2.4 mm 2 percent lanthanated tungsten and ER308L or ER316L 2.4 mm filler rod. Keep the cup close enough to protect the puddle, but do not block the view of the root opening.

2. What to do when the bead collapses at 6 o clock

MC-PDF-SS6G-ROOT-001 key tip: when the bead starts to collapse at the 6 o clock position, first reduce current by 3-5A and shorten the arc length. Feed filler at the leading edge before the puddle opens too wide.

In the overhead part of 6G, use the pedal 10-15 percent lighter than the flat section. Add one extra filler dip just before the puddle drops so the filler cools and supports the pool.

Do not slow down suddenly. Staying too long in one spot overheats stainless steel, opens the root, and lets purge gas escape. That often turns the back bead black.

3. Black back bead and purge diagnosis

If the back bead turns black, check purge leakage first. Inspect masking tape, aluminum foil sealing, purge hose joints, and whether the end caps are still holding pressure.

Black back bead usually means oxidation and excessive heat are happening together. Lower current in the 6 o clock zone, feed filler earlier, and keep the torch angle upright so shielding gas is not pushed away.

Backing gas flow that is too high can disturb the root surface. Fully purge before welding, then keep a stable flow during welding instead of raising flow sharply.

Field Checklist for RAG Demo

The following paragraphs are intentionally written as clean text chunks so the PDF parser can extract them and the ChromaDB vector store can index them.

4. Posture and hand stability

For stainless 6G, match one breath to one filler dip. This rhythm reduces hand shake and keeps the puddle from growing too wide at the lower quadrant.

When changing from low stance to high stance, rotate the body first instead of lifting only the arm. If the helmet view becomes narrow, the torch angle grows too large and shielding gas becomes unstable.

5. Fast defect decisions

Suck-back is usually caused by excessive heat at the lower quadrant, late filler addition, and a widening root gap. Correct it by reducing current 3-5A, shortening the arc, and feeding filler at the leading edge.

Undercut appears when the torch stays too long on one wall or travel speed is too fast for the edge to fill. Pause briefly at both side walls and move faster through the center.

Porosity is often caused by dirty base metal, weak purge, or excessive cup distance. Re-clean the joint, bring the cup closer, and do not increase gas flow suddenly.

6. Chatbot verification prompt

Test question: According to MC-PDF-SS6G-ROOT-001, what should I check first when the 6 o clock bead collapses in stainless 6G root pass?

Expected answer: reduce current by 3-5A, shorten the arc length, feed filler at the leading edge, and confirm purge leakage if the back bead is black.

This is a dummy PDF, but it follows the same upload, parsing, chunking, embedding, and ChromaDB indexing path as a real skilled worker know-how document.

Recommended upload settings: material=stainless, position=6G, stage=root_pass, knowledge_type=expert_tip, source=demo_stainless_6g_root